



Exercise Sheet 09

Published: July 14, 2025

Due: July 21, 2025

Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. Information Theory and Data Compression

- (a) Demonstrate the additivity of the information measure by showing that the information gained from observing the collective outcome of N independent events, each with probabilities p_i for $i = 1$ to N , is equal to the sum of the information gained from observing each event individually and independently, regardless of the order in which they are observed.
(Hint: The information gain measure assigns $\log_2(p)$ bits to an event with a probability of p .)

3P

2. Graphs and Description Length Minimization

- (a) Consider a graph $G = (V, E)$ with the following adjacency matrix:

$$\mathbb{A} = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Assuming that the row and column numbers correspond to nodes (enumerated from 1 to 5), calculate the probability that a random walker starting in node 1 will traverse the following sequence of nodes: (1, 2, 3, 2, 4, 1, 5, 1, 2).

(Hint: Consider $p(1, 2, 3, 2, 4, 1, 5, 1, 2|1)$.)

- (b) Show that for an undirected network, the stationary visitation probability of a node v converges to $\pi_v = \frac{d(v)}{2|E|}$.

2P

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3. MapEquation

- (a) Consider the following undirected network $G = (V, E)$ and a mapping $C_2 : V \rightarrow \{1, 2\}$ of nodes into two communities (red and blue). Use the MapEquation to calculate the minimal code length $L(C_2)$.

(Hint: In this case, assume that a self-loop only contributes 1 to the degree of the node)

3P

